

Summer Bootcamp – Session 1

Corrections to the Linear Algebra Exercises

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Correction Guidelines

The computations below use the column-vector convention of the exercise sheet. For numerical eigendecompositions, an eigenvector and its negative are equivalent; small differences in the last displayed decimal are due to rounding.

1 Core Linear Algebra

Exercise 1 – Matrix arithmetic and dimension checks. The dimensions are $A : 2 \times 3$, $B : 3 \times 2$, $C : 2 \times 2$, $x : 3 \times 1$, and $y : 2 \times 1$. Hence

Expression	AB	BA	AC	CA	Ax	Ay	Bx	By	C^2	$A^\top C$
Defined?	✓	✓	×	✓	✓	×	×	✓	✓	✓
Dimension	2×2	3×3	–	2×3	2×1	–	–	3×1	2×2	3×2

The requested products are

$$AB = \begin{pmatrix} -3 & -1 \\ 9 & 8 \end{pmatrix}, \quad BA = \begin{pmatrix} 2 & 7 & 2 \\ -1 & -2 & 1 \\ 3 & 12 & 5 \end{pmatrix},$$

$$Ax = \begin{pmatrix} -3 \\ -6 \end{pmatrix}, \quad By = \begin{pmatrix} 7 \\ -3 \\ 11 \end{pmatrix}, \quad C^2 = \begin{pmatrix} 3 & -3 \\ 3 & 0 \end{pmatrix}.$$

Moreover,

$$A^\top = \begin{pmatrix} 1 & 0 \\ 2 & 3 \\ -1 & 4 \end{pmatrix}, \quad B^\top = \begin{pmatrix} 2 & -1 & 3 \\ 1 & 0 & 2 \end{pmatrix},$$

$$\operatorname{tr}(C) = 3, \quad \det(C) = 3, \quad C^{-1} = \frac{1}{3} \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}.$$

Since AB is 2×2 whereas BA is 3×3 , they cannot be equal; this explicitly shows that multiplication is not commutative. Finally, if a_1, a_2, a_3 are the columns of A , then

$$Ax = a_1 - 2a_2 + 0a_3 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} - 2 \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} -3 \\ -6 \end{pmatrix}.$$

Exercise 2 – Kernel, image, rank, and geometry. The equations $Lx = 0$ imply $x_1 = x_2 = x_3$. Therefore

$$\ker(L) = \operatorname{span}\{(1, 1, 1)^\top\}, \quad \dim \ker(L) = 1, \quad \operatorname{rank}(L) = 3 - 1 = 2.$$

Every column has entries summing to zero, hence is orthogonal to $\mathbf{1} = (1, 1, 1)^\top$. Thus $\operatorname{Im}(L) \subset \mathbf{1}^\perp$. Both spaces have dimension two, so

$$\operatorname{Im}(L) = \mathbf{1}^\perp = \{y \in \mathbb{R}^3 : y_1 + y_2 + y_3 = 0\}.$$

The vector b_1 has coordinate sum zero, so solutions exist. One is $x_0 = (1, 0, -1)^\top$, and all solutions are

$$x = (1, 0, -1)^\top + t(1, 1, 1)^\top, \quad t \in \mathbb{R}.$$

The coordinates of b_2 sum to two, so $b_2 \notin \text{Im}(L)$ and no solution exists. Geometrically, L kills the direction $\text{span}\{\mathbf{1}\}$ and acts non-trivially on the plane $\mathbf{1}^\perp$.

Exercise 3 – Linear systems and bootstrapping. Part A. Here

$$A = \begin{pmatrix} 2 & 3 \\ 1 & -4 \end{pmatrix}, \quad b = \begin{pmatrix} 5 \\ -2 \end{pmatrix}, \quad \det(A) = -11 \neq 0.$$

The solution is unique. From $x_1 = -2 + 4x_2$, substitution gives $11x_2 = 9$, so

$$x_2 = \frac{9}{11}, \quad x_1 = \frac{14}{11}.$$

Also

$$A^{-1} = \begin{pmatrix} 4/11 & 3/11 \\ 1/11 & -2/11 \end{pmatrix}, \quad A^{-1}b = \begin{pmatrix} 14/11 \\ 9/11 \end{pmatrix}.$$

Part B. The pricing system is

$$\underbrace{\begin{pmatrix} 1.03 & 0 & 0 \\ 0.04 & 1.04 & 0 \\ 0.05 & 0.05 & 1.05 \end{pmatrix}}_C \begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix} = \begin{pmatrix} 1.005 \\ 1.018 \\ 1.025 \end{pmatrix}.$$

It is lower triangular because a bond cannot pay after its maturity. Forward substitution yields

$$d_1 = \frac{1.005}{1.03} = 0.975728,$$

$$d_2 = \frac{1.018 - 0.04d_1}{1.04} = 0.941318, \quad d_3 = \frac{1.025 - 0.05d_1 - 0.05d_2}{1.05} = 0.884903.$$

Consequently,

$$(z_1, z_2, z_3) = (0.024571, 0.030237, 0.040759),$$

i.e. approximately (2.457%, 3.024%, 4.076%). Each maturity is obtained using discount factors already computed at earlier maturities: this sequential triangular solve is exactly bootstrapping.

Exercise 4 – Covariance matrices and portfolio variance. Both columns sum to zero. Direct multiplication gives

$$X^\top X = \begin{pmatrix} 0.0010 & -0.0002 \\ -0.0002 & 0.0006 \end{pmatrix}, \quad \hat{\Sigma} = \begin{pmatrix} 0.000333333 & -0.000066667 \\ -0.000066667 & 0.000200000 \end{pmatrix}.$$

The off-diagonal entries coincide, so the matrix is symmetric. For $w = (0.6, 0.4)^\top$,

$$w^\top \hat{\Sigma} w = 0.000120.$$

Indeed,

$$0.6^2(0.000333333) + 0.4^2(0.0002) + 2(0.6)(0.4)(-0.000066667) = 0.000120.$$

A positive covariance increases the cross term; a negative covariance reduces it and creates diversification, subject to the covariance matrix remaining positive semidefinite.

Exercise 5 – Diagonalization and covariance interpretation. The symmetry of the two coordinates suggests

$$q_+ = \frac{1}{\sqrt{2}}(1, 1)^\top, \quad q_- = \frac{1}{\sqrt{2}}(1, -1)^\top.$$

Then

$$C_\rho q_+ = (1 + \rho)q_+, \quad C_\rho q_- = (1 - \rho)q_-.$$

For $\rho = 0.8$,

$$Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad D = \text{diag}(1.8, 0.2), \quad C_\rho = Q D Q^\top.$$

The first vector is a common long–long direction; the second is a spread or long–short direction. When $\rho \rightarrow 1$, the spread eigenvalue $1 - \rho$ becomes small. At $\rho = -1$, the eigenvalues are 0 and 2; the common direction has zero variance and the matrix is singular.

Exercise 6 – PCA by hand. Since $n = 3$,

$$\hat{\Sigma} = \frac{1}{2} X^\top X = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Its eigenpairs are

$$\lambda_1 = 2, \quad q_1 = \frac{1}{\sqrt{2}}(1, 1)^\top, \quad \lambda_2 = 0, \quad q_2 = \frac{1}{\sqrt{2}}(1, -1)^\top.$$

The first component scores are

$$X q_1 = (-\sqrt{2}, 0, \sqrt{2})^\top.$$

Reconstruction using one component is

$$(X q_1) q_1^\top = \begin{pmatrix} -1 & -1 \\ 0 & 0 \\ 1 & 1 \end{pmatrix} = X.$$

It is exact because the data have rank one. The common direction is retained; the orthogonal spread direction contains zero sample variation and nothing is lost.

2 Hands-On Python

Exercise 7 – Empirical covariance and PCA with NumPy. One complete implementation is:

```
mu = R.mean(axis=0)
Rc = R - mu
Sigma = Rc.T @ Rc / (len(R) - 1)
assert np.allclose(Sigma, np.cov(R, rowvar=False))

eigvals, eigvecs = np.linalg.eigh(Sigma)
idx = np.argsort(eigvals)[::-1]
eigvals, eigvecs = eigvals[idx], eigvecs[:, idx]
ratios = eigvals / eigvals.sum()
Z1 = Rc @ eigvecs[:, :1]
Z2 = Rc @ eigvecs[:, :2]
```

The numerical results are

$$\mu = (0.002375, 0.000375, 0.002000, 0.003000)^\top,$$

$$\hat{\Sigma} = \begin{pmatrix} 0.000100839 & 0.000011982 & 0.000059143 & 0.000109571 \\ 0.000011982 & 0.000015696 & 0.000008714 & 0.000009286 \\ 0.000059143 & 0.000008714 & 0.000035429 & 0.000063429 \\ 0.000109571 & 0.000009286 & 0.000063429 & 0.000121143 \end{pmatrix}.$$

In decreasing order, the eigenvalues and explained-variance ratios are

$$\lambda = (2.56457 \times 10^{-4}, 1.5480 \times 10^{-5}, 1.010 \times 10^{-6}, 1.59 \times 10^{-7}),$$

$$r = (0.93904, 0.05668, 0.00370, 0.00058).$$

One possible first eigenvector is

$$q_1 = (-0.6265, -0.0708, -0.3667, -0.6841)^\top.$$

The one-component scores are approximately

$$Z_1 = (-0.01209, 0.01405, -0.01847, 0.02165, -0.00899, -0.00054, 0.01879, -0.01441)^\top.$$

All entries of q_1 have the same sign, so the first component is a broad market-like factor. Its sign is arbitrary; replacing q_1 by $-q_1$ changes the scores' signs but not the PCA.

Exercise 8 – Portfolio risk lab. The matrix is symmetric and

$$\text{eigvalsh}(\Sigma) = (0.032725, 0.050366, 0.089182, 0.177727),$$

so it is positive definite. For the equal-weight portfolio,

$$\mathbb{E}[R_{w_0}] = 0.0825, \quad \text{Var}(R_{w_0}) = 0.035125, \quad \sigma_{w_0} = 0.187417.$$

The minimum-variance weights are obtained without forming the inverse:

```
ones = np.ones(4)
v = np.linalg.solve(Sigma, ones)
w_star = v / (ones @ v)
```

They are

$$w^* = (0.520739, 0.288460, 0.102549, 0.088252)^\top.$$

Their return, variance, and volatility are

$$0.066049, \quad 0.0270195, \quad 0.164376.$$

A reproducible random comparison is:

```
rng = np.random.default_rng(12345)
W = rng.random((5000, 4))
W /= W.sum(axis=1, keepdims=True)
vol = np.sqrt(np.einsum("ij,jk,ik->i", W, Sigma, W))
i = np.argmin(vol)
print(W[i], vol[i])
```

This gives weights approximately (0.5128, 0.3110, 0.0950, 0.0811) and volatility 0.164488. It is close but not identical: random search samples finitely many points, whereas the analytic solution finds the exact continuous optimum.

3 Decomposition Brainteasers and Applications

Exercise 9 – Predicting a coupled system without iterating. Direct calculation gives

$$x_1 = (1, 2)^\top, \quad x_2 = (5, 4)^\top, \quad x_3 = (19, 8)^\top.$$

The eigenvalues are 3 and 2, with

$$P = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \quad D = \text{diag}(3, 2), \quad P^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Therefore

$$A^k = P D^k P^{-1} = \begin{pmatrix} 3^k & 3^k - 2^k \\ 0 & 2^k \end{pmatrix}, \quad x_k = \begin{pmatrix} 3^k - 2^k \\ 2^k \end{pmatrix}.$$

The eigenvalue 3 controls generic long-run growth. If $x_0 = (-1, 1)^\top$, it is already an eigenvector for eigenvalue 2, so $x_k = 2^k(-1, 1)^\top$ and the 3^k mode vanishes.

Exercise 10 – Finding hidden risk factors. The candidates $(1, 1)^\top$ and $(1, -1)^\top$ are orthogonal. After normalization they are q_+ and q_- from Exercise 5, with variances/eigenvalues 1.8 and 0.2. The first is a common trade; the second is a spread trade. The equally weighted long–short portfolio $(1/2, -1/2)^\top$ has variance

$$(1/2, -1/2)C(1/2, -1/2)^\top = 0.1,$$

so it is low risk but not riskless. In general the common and spread variances are $1 + \rho$ and $1 - \rho$. As $\rho \rightarrow 1$ the spread becomes riskless; as $\rho \rightarrow -1$ the common direction becomes riskless.

Exercise 11 – Manufacturing a target correlation. Matching entries in $LL^\top = \Sigma$ gives

$$L = \begin{pmatrix} 2 & 0 \\ 1 & \sqrt{2} \end{pmatrix}.$$

Thus

$$X_1 = 2Z_1, \quad X_2 = Z_1 + \sqrt{2}Z_2.$$

Independence gives

$$\text{Var}(X_1) = 4, \quad \text{Var}(X_2) = 1 + 2 = 3, \quad \text{Cov}(X_1, X_2) = 2,$$

and hence $\rho_{12} = 2/\sqrt{12} = 1/\sqrt{3}$. For $z = (1, -1)^\top$,

$$Lz = (2, 1 - \sqrt{2})^\top.$$

Multiplying Z by Σ would produce covariance $\Sigma I \Sigma^\top = \Sigma^2$, not Σ .

4 Finance Brainteasers and Theory

Exercise 12 – Singular covariance and redundant assets. First use bilinearity of covariance:

$$\text{Cov}(R_1, R_3) = \text{Cov}(R_1, R_1 + R_2) = \text{Var}(R_1) + \text{Cov}(R_1, R_2) = 1.2,$$

$$\text{Cov}(R_2, R_3) = \text{Cov}(R_2, R_1) + \text{Var}(R_2) = 1.2,$$

and

$$\text{Var}(R_3) = \text{Var}(R_1 + R_2) = \text{Var}(R_1) + \text{Var}(R_2) + 2 \text{Cov}(R_1, R_2) = 2.4.$$

Therefore

$$\Sigma = \begin{pmatrix} 1 & 0.2 & 1.2 \\ 0.2 & 1 & 1.2 \\ 1.2 & 1.2 & 2.4 \end{pmatrix}.$$

There is also a structural derivation. Set

$$X = \begin{pmatrix} R_1 \\ R_2 \end{pmatrix}, \quad M = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

Then $R = (R_1, R_2, R_3)^\top = MX$, so

$$\Sigma = \text{Cov}(R) = M \text{Cov}(X) M^\top = M \begin{pmatrix} 1 & 0.2 \\ 0.2 & 1 \end{pmatrix} M^\top.$$

This factorization already implies $\text{rank}(\Sigma) \leq \text{rank}(M) = 2$.

Take $w = (-1, -1, 1)^\top$. Direct multiplication gives

$$\Sigma w = \begin{pmatrix} -1 - 0.2 + 1.2 \\ -0.2 - 1 + 1.2 \\ -1.2 - 1.2 + 2.4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

The corresponding portfolio return is

$$w^\top R = -R_1 - R_2 + R_3 = 0$$

in every outcome, not merely in expectation.

The singularity has three equivalent explanations:

- the returns satisfy the exact relation $R_3 - R_1 - R_2 = 0$;
- the covariance matrix has the nonzero kernel vector w ;
- $R = MX$ depends on only two underlying random returns and $\text{rank}(M) = 2 < 3$.

Moreover, $\text{Cov}(X)$ has eigenvalues 1.2 and 0.8, so it is positive definite. Hence $\text{rank}(\Sigma) = 2$ and

$$\ker(\Sigma) = \text{span}\{(-1, -1, 1)^\top\}.$$

Let $p = (p_1, p_2, p_3)^\top$ be the current price vector. Since w produces the zero payoff, no arbitrage requires it to have zero price:

$$w^\top p = -p_1 - p_2 + p_3 = 0, \quad \text{so} \quad \boxed{p_3 = p_1 + p_2}.$$

If $p_3 < p_1 + p_2$, buy one unit of security 3 and short one unit of each of securities 1 and 2. The initial inflow is $p_1 + p_2 - p_3 > 0$, while the terminal payoff is $R_3 - R_1 - R_2 = 0$.

If $p_3 > p_1 + p_2$, reverse the trade: short security 3 and buy securities 1 and 2. The initial inflow is $p_3 - p_1 - p_2 > 0$, while the terminal payoff is $R_1 + R_2 - R_3 = 0$. Thus either violation of the price relation creates a zero-risk arbitrage.

Exercise 13 – Short theoretical questions.

1. $(X^\top X)^\top = X^\top X$, and for every z , $z^\top X^\top X z = \|Xz\|^2 \geq 0$. Hence $X^\top X$ is symmetric positive semidefinite.
2. If A is injective and $Ax = 0 = A0$, then $x = 0$. Conversely, if $\ker(A) = \{0\}$ and $Ax = Ay$, then $A(x - y) = 0$, hence $x = y$.
3. The statement $b \in \text{Im}(A)$ means by definition that there exists an x such that $b = Ax$. This is exactly solvability of $Ax = b$.
4. If $Au = \lambda u$, $Av = \mu v$, and $A = A^\top$, then $\lambda u^\top v = (Au)^\top v = u^\top Av = \mu u^\top v$. If $\lambda \neq \mu$, then $u^\top v = 0$.
5. The Jordan block $J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has the repeated eigenvalue 1 but only the one-dimensional eigenspace $\text{span}\{(1, 0)^\top\}$, so it is not diagonalizable.
6. A covariance matrix is real symmetric positive semidefinite, so the spectral theorem gives $\hat{\Sigma} = QDQ^\top$. The columns of Q are orthogonal directions and the diagonal entries of D are the variances after projection. Ordering those eigenvalues from largest to smallest is precisely PCA; truncation keeps the directions explaining the most variance.

5 Markov Chains and Matrix Brainteasers

Exercise 14 – Credit migration by matrix powers. The entries are nonnegative and each column sums to one. Column G says G→G with probability 0.9 and G→B with probability 0.1; column B says B→G with probability 0.4 and B→B with probability 0.6. Starting from G,

$$\mu_1 = (0.9, 0.1)^\top, \quad \mu_2 = (0.85, 0.15)^\top.$$

Solving $P\pi = \pi$ and $\mathbf{1}^\top \pi = 1$ gives $\pi = (0.8, 0.2)^\top$. The eigenvalues are 1 and 0.5. One diagonalization is

$$P = SDS^{-1}, \quad S = \begin{pmatrix} 4 & 1 \\ 1 & -1 \end{pmatrix}, \quad D = \text{diag}(1, 0.5), \quad S^{-1} = \begin{pmatrix} 0.2 & 0.2 \\ 0.2 & -0.8 \end{pmatrix}.$$

Therefore

$$\mu_n = \begin{pmatrix} 0.8 \\ 0.2 \end{pmatrix} + 0.2(0.5)^n \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

The Bad-state error is $0.2(0.5)^n$. We need it at most 10^{-3} , or $(0.5)^n \leq 0.005$, whose smallest integer solution is $n = 8$.

Exercise 15 – Waiting time for PPF. Use states 0, P, PP, and the absorbing state PPF. They record the longest suffix of the observations that is a prefix of PPF. In the order (0, P, PP, PPF), with columns denoting the current state, the transition matrix is

$$P = \begin{pmatrix} 1/2 & 1/2 & 0 & 0 \\ 1/2 & 0 & 0 & 0 \\ 0 & 1/2 & 1/2 & 0 \\ 0 & 0 & 1/2 & 1 \end{pmatrix}.$$

The first-step recurrences are

$$\begin{aligned} E_0 &= 1 + \frac{1}{2}E_P + \frac{1}{2}E_0, \\ E_P &= 1 + \frac{1}{2}E_{PP} + \frac{1}{2}E_0, \quad E_{PP} = 1 + \frac{1}{2}E_{PP}. \end{aligned}$$

Thus $E_{PP} = 2$, $E_P = 2 + E_0/2$, and $E_0 = 2 + E_P$. Solving gives

$$\boxed{E_0 = 8}, \quad E_P = 6, \quad E_{PP} = 2.$$

From state PP, observing another P produces PPP, whose longest suffix that is a prefix of PPF is still PP. The useful partial match is therefore not lost.

Exercise 16 – Waiting time for 564. Use states 0, 5, 56, and absorbing state 564. In this order, the column-stochastic transition matrix is

$$P = \begin{pmatrix} 5/6 & 4/6 & 4/6 & 0 \\ 1/6 & 1/6 & 1/6 & 0 \\ 0 & 1/6 & 0 & 0 \\ 0 & 0 & 1/6 & 1 \end{pmatrix}.$$

The first-step equations are

$$\begin{aligned} E_0 &= 1 + \frac{1}{6}E_5 + \frac{5}{6}E_0, \\ E_5 &= 1 + \frac{1}{6}E_{56} + \frac{1}{6}E_5 + \frac{4}{6}E_0, \\ E_{56} &= 1 + \frac{1}{6}E_5 + \frac{4}{6}E_0. \end{aligned}$$

The first equation gives $E_0 = 6 + E_5$. Solving the system yields

$$\boxed{E_0 = 216}, \quad E_5 = 210, \quad E_{56} = 180.$$

The answer equals $6^3 = 216$. This equality holds because the word 564 has no nontrivial overlap with itself: after a failed completion, no prefix longer than the explicitly tracked suffix survives. For patterns with self-overlap, the expected waiting time need not equal the reciprocal 6^3 of the single-block probability.